

Project Pitch

Project Title: Yatra — AI-Powered Personalized EdTech Platform

Tagline: Intelligent educational journey powered by fast inference LLMs and modern full-stack architecture.

Description

Problem

Traditional EdTech platforms deliver static content with one-size-fits-all curricula. This leads to low engagement, high dropout rates, and inefficient learning. Students learn at different paces and styles, while educators lack tools to rapidly create, adapt, and scale high-quality interactive content.

Solution

Yatra (Sanskrit for "journey") is a full-stack AI-native EdTech platform that delivers **truly personalized learning experiences** using real-time LLM inference.

Built with a modern, scalable architecture, Yatra combines a performant Next.js frontend with a FastAPI backend, already integrated with **Groq** for ultra-fast AI interactions.

Technical Implementation (Current State)

Frontend:

- Next.js 16 (App Router) + React 18 + TypeScript
- Tailwind CSS + Framer Motion for smooth, responsive UI/UX
- Server Components + Client Components architecture
- Groq SDK integration for low-latency AI features

Backend:

- FastAPI (Python) with Uvicorn ASGI server
- SQLAlchemy 2.0 + Pydantic v2 for data modeling
- Clean separation of concerns, ready for PostgreSQL/Supabase
- RESTful API structure designed for future real-time extensions (WebSockets)

Key Technical Highlights:

- Monorepo structure (frontend + backend)
- Modular and extensible architecture
- Ready for authentication via Supabase Auth / NextAuth
- Optimized for fast iteration during hackathon development
- Groq integration enables sub-second responses for chat-based tutoring, lesson generation, and exercise creation

Core AI Features (In Progress / Planned for Demo)

- Real-time AI tutor using Groq's Llama-3 / Mixtral models
- Dynamic lesson & quiz generation based on student's knowledge level
- Adaptive learning paths with progress tracking
- Automated exercise evaluation and feedback
- AI-assisted content creation tools for educators

Architecture Overview

- **Client Layer:** Next.js 16 App Router (SSR + CSR hybrid)
- **AI Layer:** Groq Cloud API (fast inference)
- **Backend Layer:** FastAPI + SQLAlchemy ORM
- **Data Layer:** PostgreSQL / Supabase (planned)
- **Styling & UX:** Tailwind + Framer Motion

Target Audience

- University students and lifelong learners
- Educators and course creators
- Educational institutions looking for modern AI tools

Future Roadmap

- Real-time collaboration and peer-learning features
- Advanced analytics dashboard with learning insights
- Multi-model support (Groq + other providers)
- Mobile PWA / React Native version
- LMS integration (Moodle, Canvas, etc.)

Why Yatra stands out: We are not just wrapping ChatGPT in a nice UI. We are building a complete educational infrastructure with modern development practices, high-performance AI inference, and thoughtful UX designed specifically for learning outcomes.

Slogan: *Yatra — Because learning is a journey, not a checklist.*